

Prevalence of *Yersinia enterocolitica* shedding and bioserotype distribution in Ontario finisher pig herds in 2001, 2002, and 2004.

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We investigated characteristics of *Yersinia enterocolitica* infection in Ontario finisher pig herds. Our specific objectives were to estimate or test: prevalence of *Y. enterocolitica* shedding in finisher pigs, bioserotype distribution, agreement between the herd-level tests based on sampling pig and pooled fecal samples, whether bioserotypes cluster by farms, and whether *Y. enterocolitica*-positive herds cluster spatially. In total, 3747 fecal samples were collected from 100 farms over the years 2001, 2002, and 2004 (250 total herd visits). Fecal samples were tested by culture and positive isolates were biotyped and serotyped. Apparent pig-level prevalence of *Y. enterocolitica* was 1.8%, 3.2%, and 12.5% in 2001, 2002, and 2004, respectively. Estimated true pig-level prevalence of *Y. enterocolitica* was 5.1%, 9.1%, and 35.1% in 2001, 2002, and 2004, respectively. Herd-level prevalence was 16.3%, 17.9%, and 37.5% in 2001, 2002, and 2004, respectively. In all years, the most common bioserotype was 4, O:3, followed by bioserotype 2, O:5,27. Kappa between herd-level status based on pig and pooled samples ranged between 0.51 and 0.68 for biotype 1A and bioserotype 4, O:3, respectively. For 4, O:3, a significant bias in discordant pairs was detected, indicating that pig samples were more sensitive than pooled samples in declaring a herd as positive. Farms tended to be repeatedly positive with the same bioserotype, but positive study farms did not cluster spatially (suggesting lack of between herd transmission and lack of a common geographic risk factor). Copyright 2009 Elsevier B.V. All rights reserved.